

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with nine years delivering production-grade applications across financial services and SaaS environments. Deep experience in TypeScript, React, and distributed backend systems including REST and GraphQL APIs, event-driven pipelines, and PostgreSQL schema design. Known for end-to-end ownership spanning automated testing, observability, on-call operations, and cross-team architectural guidance that improves platform reliability and performance at scale.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led development of a **TypeScript and React** portfolio analytics platform serving investment teams with low-latency views into positions, cash flows, and payout workflows across high-volume daily usage.
- Designed and operated **REST and GraphQL APIs** that power account activity, entitlement checks, and trader-facing dashboards while enforcing authentication, authorization, and audit logging at the service boundary.
- Architected **event-driven processing pipelines** with message queues to decouple trade ingestion, notification delivery, and reporting jobs, applying idempotency and retry patterns to manage race conditions and eventual consistency.
- Owned **PostgreSQL schema design**, migrations, and indexing strategy for core ledger and activity tables, improving query plans on portfolio history reads and reducing operational toil during schema evolution.
- Embedded **automated testing and safe deployment practices** into feature delivery through integration tests, contract checks, and staged rollouts that limited blast radius during frequent production releases.
- Instrumented services with **metrics, structured logging, and distributed tracing**, enabling faster triage during on-call rotations and clearer visibility into latency and error budgets for customer-facing endpoints.
- Partnered with product and operations stakeholders to translate complex trading workflow requirements into **cloud-native services** with clear SLIs, runbooks, and operational ownership from design through production support.
- Mentored engineers on **API design, React state management, and database migration safety**, elevating code review quality and helping newer contributors ship independently on shared platform modules.
- Influenced **engineering standards and reusable UI patterns** adopted across product squads, improving consistency of frontend architecture and accelerating delivery of new trader experience features.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Built **internal risk and reporting applications** used by trading and operations teams, connecting Java services to dashboards that surfaced exposure, P&L, and settlement status in near real time.
- Developed **REST APIs** that integrated legacy banking platforms with modern front-end tools, standardizing payload contracts and error handling for downstream consumer teams.
- Optimized **SQL Server queries, indexes, and stored procedures** supporting regulatory and management reporting over large transaction histories without degrading batch completion windows.
- Contributed to **event-driven notification services** that propagated market and booking updates to analytics consumers, incorporating deduplication and backoff logic for unreliable network paths.
- Supported **production incident response** for critical batch and API services, documenting root causes and remediation steps that strengthened recovery playbooks for recurring failure modes.
- Collaborated with compliance and business partners to deliver **secure, auditable workflows** for trade capture and reference data changes aligned with bank control requirements.

- Automated **CI/CD pipelines** for microservice deployments across QA and production environments, reducing manual release steps and improving repeatability of environment promotion.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Shipped **React and Node.js features** for hospitality compliance software, enabling clients to complete inspection checklists and review audit history from web and tablet interfaces.
- Implemented **REST endpoints and PostgreSQL models** for checklist submissions, user sessions, and activity logs with validation rules that kept data consistent across concurrent updates.
- Added **automated test coverage** for authentication flows and core API handlers, catching regressions early in the release cycle for modules touched by frequent product iterations.
- Improved **dashboard performance** by refining component rendering paths and paginating large result sets, making daily operational views more responsive for field users.
- Participated in code reviews and paired sessions that accelerated adoption of **TypeScript** across shared frontend modules while preserving backward compatibility.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive fundraising and program pages** using HTML, CSS, JavaScript, and WordPress, delivering polished public sites aligned with nonprofit branding guidelines.
- Integrated **donation forms and analytics tracking** so stakeholders could monitor campaign traffic and conversion activity without manual reporting overhead.
- Ensured **cross-browser compatibility and accessibility basics** on educational content pages viewed by diverse community audiences and partner organizations.
- Launched **event landing pages** on short timelines by coordinating content updates, asset delivery, and QA with volunteer marketing contributors.
- Documented **hosting and deployment steps** for foundation staff, enabling smoother maintenance and handoff after the engagement ended.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Redux, Component Architecture, Responsive UI

Backend: Node.js, GraphQL, REST API Design, Java, Spring Boot, Event-Driven Architecture, Message Queues, Microservices

Databases: PostgreSQL, SQL Server, Schema Design, Migrations, Indexing, Query Optimization, Stored Procedures

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Terraform, Cloud-Native Deployments, Infrastructure as Code

Monitoring & Tools: Datadog, Prometheus, Grafana, OpenTelemetry, Git, Jest, Cypress, Postman

Workflow & PM: Agile/Scrum, Jira, Confluence, Cross-Functional Collaboration, Technical Mentorship, On-Call Operations